The apparatus shown consists of three temperature jacketed 1.6 L bulbs connected by stopcocks. Bulb A contains a mixture of $H_2O(g)$, $CO_2(g)$ and $N_2(g)$ at 27°C and a total pressure of 570 mm Hg. Bulb B is empty and held at a temperature of −73°C. Bulb C is also empty and is held at a temperature of −193°C. The stopcocks are closed and the volume of the lines connecting the bulbs is zero. CO_2 sublimes at −78°C and N_2 boils at −196°C. Neglect the volume of any solid, if formed. (Take R = 0.08 lit atm mol^{-1} K^{-1})

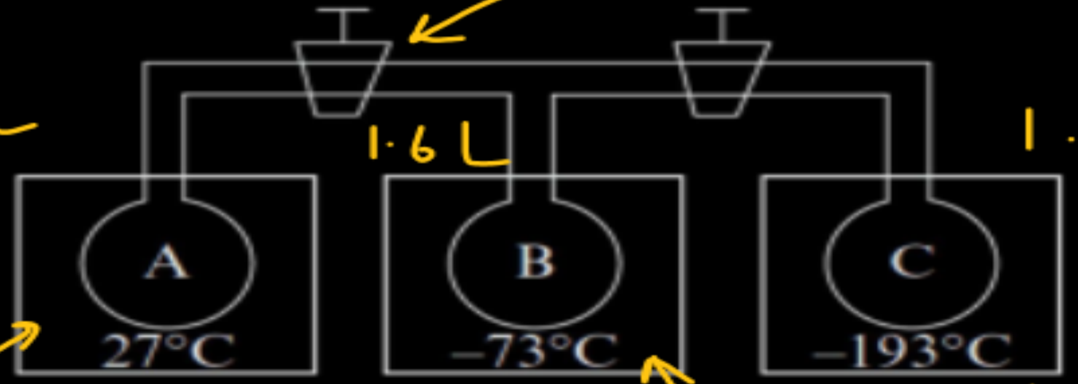
when both valves opened, A $\longrightarrow N_2$

B $\longrightarrow H_2O(s), N_2$

C $\longrightarrow H_2O(s), CO_2(s) + N_2(g)$.

The stopcock between A and B is opened and the system is allowed to come to equilibrium. The pressure in A and B is now 0.21 atm. What do bulbs A and B contain?

570 torr



1.6L 1.6L 1.6L

300K 200K 80K

273 − 193 = 80

(A) A: $CO_2(g)$,
B: $N_2(g)$, $H_2O(s)$

(B) A: $N_2(g)$,
B: $CO_2(g)$, $N_2(g)$, $H_2O(s)$

(C) A: $CO_2(g)$, $N_2(g)$,
B: $CO_2(g)$, $N_2(g)$, $H_2O(s)$

(D) A: $CO_2(g)$, $N_2(g)$, $H_2O(l)$,
B: $CO_2(g)$, $N_2(g)$

$$\frac{570}{760} \times 1.6 = (n_{H_2O} + n_{N_2} + n_{CO_2}) \times 0.08 \times 300$$

$$n_{H_2O} + n_{N_2} + n_{CO_2} = \frac{57 \times 1.6}{76 \times 8 \times 30} = \frac{1}{20} = 0.05 \text{ mol}$$

$$0.21 \times V = n \times RT$$

$$(n_{N_2} + n_{CO_2}) = \frac{0.21 \times 1.6}{0.08 \times 300} + \frac{0.21 \times 1.6}{0.08 \times 200} = \frac{0.42}{30} + \frac{0.42}{20} = 0.42\left(\frac{2+3}{60}\right) = 0.035 \text{ mol}$$

$$n_{H_2O} = 0.050 - 0.035 = 0.015 \text{ mol}$$

How many moles of H_2O are in the system?

(A) 0.0075 (B) 0.015 (C) 0.05 (D) 0.035

Both stopcocks are opened and the system is again allowed to come at equilibrium. The pressure throughout the system is 22.8 mm Hg. What do bulbs A, B and C contain ?

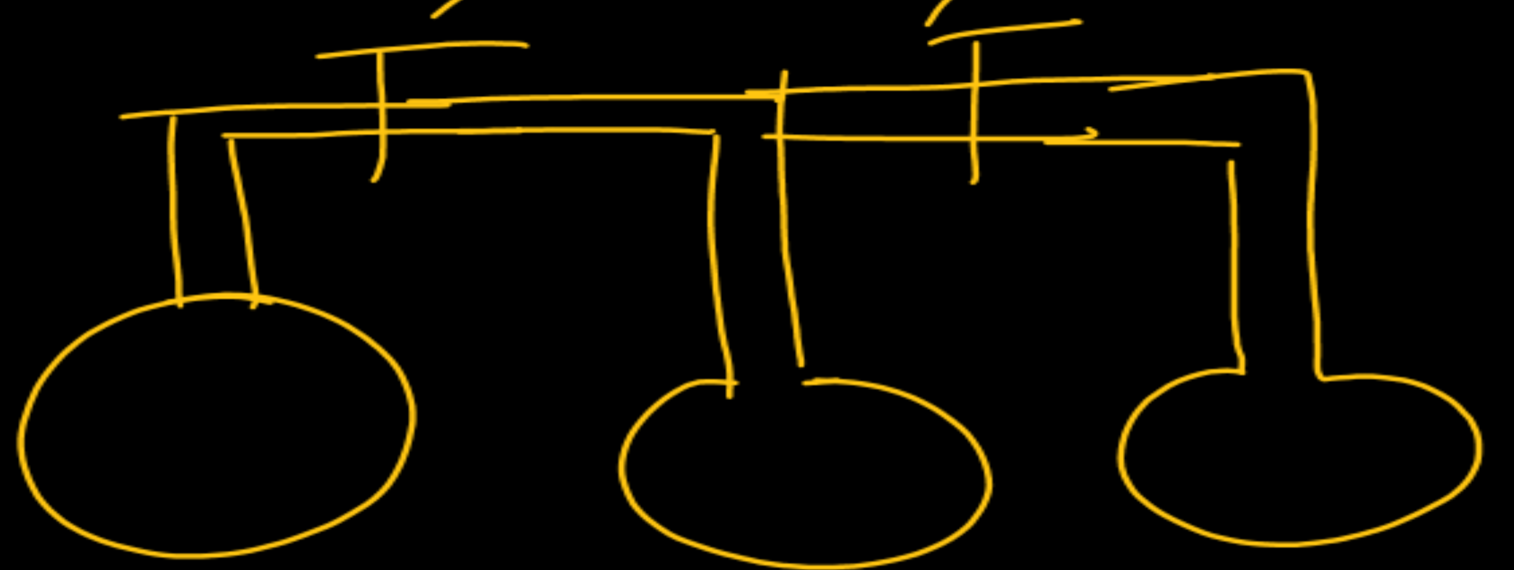
(a) A: $N_2(g)$,
B: $N_2(g)$, $H_2O(s)$,
C: $N_2(g)$, $H_2O(s)$, $CO_2(s)$

(b) A: $N_2(g)$,
B: $N_2(g)$, $H_2O(s)$,
C: $N_2(g)$, $CO_2(s)$

(c) A: $N_2(g)$,
B: $N_2(g)$, $H_2O(s)$, $CO_2(s)$,
C: $N_2(g)$, $H_2O(s)$, $CO_2(s)$

(d) A: $N_2(g)$,
B: $N_2(g)$, $H_2O(s)$, $CO_2(g)$,
C: $N_2(g)$, $H_2O(s)$, $CO_2(s)$

both opened.



N_2, CO_2, H_2O

$n_{CO_2} = 0.0350 - 0.0125$
$= 0.0225$ mol.

$PV = nRT$

$$n_{N_2} = \frac{22.8}{760} \times \frac{1}{0.08}\left[\frac{1}{300} + \frac{1}{200} + \frac{1}{80}\right]$$

$$= 0.6\left[\frac{200 \times 80 + 300 \times 80 + 300 \times 200}{300 \times 200 \times 80}\right]$$

$$= 0.6 \times \left[\frac{16 + 24 + 60}{300 \times 16}\right]$$

$$= \frac{0.6 \times 1}{3 \times 160} = \frac{1}{80}$$

$= 0.0125$ mol.

How many moles of CO_2 are in the system?

(a) 0.0125 (b) 0.015 (c) 0.0225 (d) 0.0375

The vapour pressure of water at 353 K is 532 mm Hg. The external pressure on H_2O (l) taken in a cylinder fitted with frictionless movable piston initially containing 0.9 L (= 0.9 kg) of H_2O (l) at 353 K is increased to 1 atm at the constant temperature. Then, heat is supplied keeping the pressure constant till 0.45 L of H_2O (l) is evaporated to form H_2O (g) at 373 K. Assume the internal energy of liquid to be dependent only on temperature. Answer the following questions by carefully observing the diagrams and the data provided.

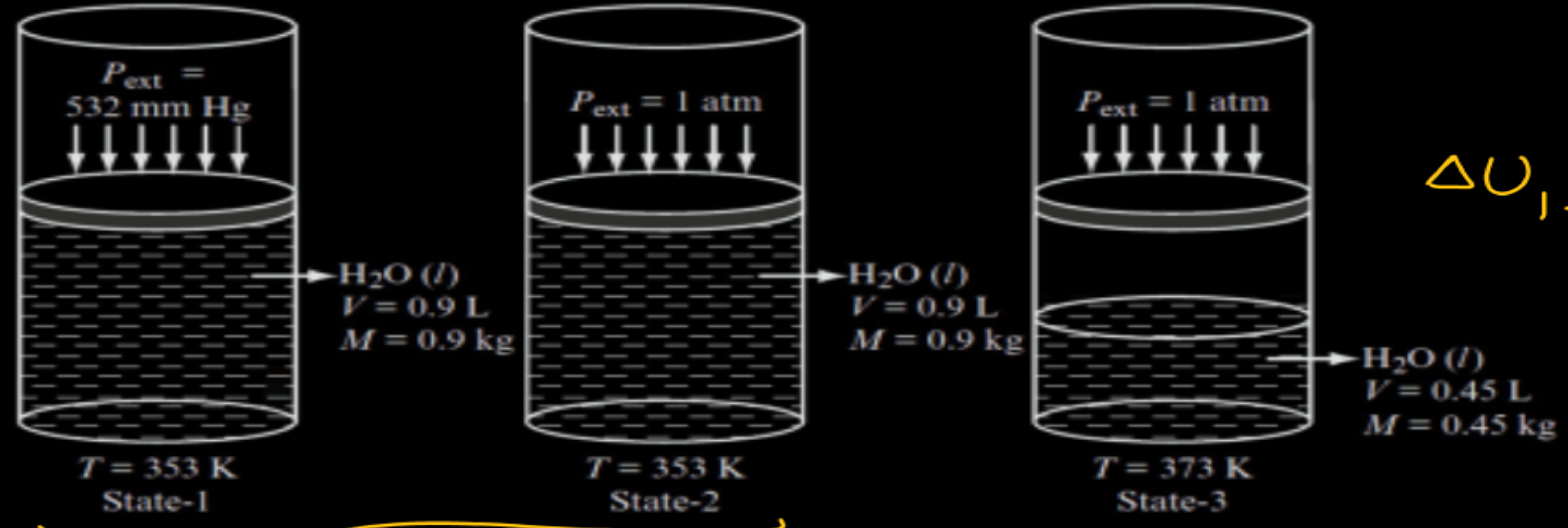
Specific heat of H_2O (l) = 4.2 J/°C-g

ΔH_{vap} at 373 K and 1 atm = +40 kJ/mol

1 L - atm = 100 J

R = 8 J/K-mol

heat added



T → Const

$$760 - 532 = 228$$

$$\Delta U = \Delta H - \Delta(PV)$$

$$= 75.6 + 1000 - 74.6 = \boxed{1001 \text{ kJ}}$$

$U = f(T)$ for liq.

$$\Delta U_{1\to2} = 0.$$

$$\Delta H_{1\to2} = \cancel{\Delta U_{1-2}}^{\text{zero}} + \Delta(PV)$$

$$= 0 + 0.9\left(1 - \frac{532}{760}\right)$$

$$= 0.9 \times \frac{\cancel{228}\ 0.3}{\cancel{760}} = 0.27 \text{ lit atm}$$

$$= 0.27 \times 100$$

$$= 27 \text{ Joule.}$$

$$\Delta U_{1\to3} = \cancel{\Delta U_{1\to2}}^{\text{zero.}} + \Delta U_{2\to3}$$

$$= 900 \times \frac{4.2}{1000} \times (373-353) + \Delta H - \Delta(PV)$$

$$= 0.9 \times 4.2 \times 20 + \frac{450}{18} \times 40 - \Delta n_g \times RT$$

$$= 18 \times 4.2 + 1000 - 25 \times 8 \times \frac{373}{1000}$$

ΔH (in J), when system is taken from State-1 to State-2, is

(a) zero (b) 0.27 (c) 27 (d) 90

Total change in ΔU (in kJ) going from State-1 to State-3 is

(a) 75.6 (b) 1075.6 (c) 1001 (d) 74.6

Total change in enthalpy (in kJ) going from State-1 to State-3 is

(a) 75.6 (b) 1075.6 (c) 1001 (d) 74.6

$\Delta H_T = \Delta H_{1\to 2} + \Delta H_{2\to 3}$ (27 J)

$= \frac{27}{1000} + 900 \times \frac{4.2}{1000} \times (373-353) + \frac{\cancel{450}^{25}}{\cancel{18}} \times 40 = \frac{27}{1000} + 75.6 + 1000$

$= 1075.6 + 0.027$

$= 1075.627$ KJ.

What is the magnitude of work done (in J) in going State-1 to State-3?

(a) Zero (b) 74.6 (c) 90 (d) 31.5

$W_T = \cancel{W_{1\to 2}}^{0} + W_{2\to 3} = -P \times \Delta V$

(V → Constant)

$= -\Delta n_g \times RT = -\frac{450}{18} \times 0.08 \times 373$

$= -74.6$ KJ

Imp

26.2 g of $MgCl_2.2H_2O$ is mixed with 92.8 g of H_2O to form a solution of density 2.38g/ml.

Choose the correct option(s) :

Ans : A, B, D

(A) Molarity of Cl^- ion = 8 molar

(B) Molality of Cl^- ion = 4 molal

(C) Molarity of Mg^{2+} ion = 8 molar

(D) Molality of Mg^{2+} ion = 2 molal

Sol^n :

$$n_{MgCl_2.2H_2O} = \frac{26.2}{24+71+36} = \frac{\cancel{26.2}}{\cancel{131}} = 0.2 \text{ mol}.$$

$$n_{Mg^{2+}} = 0.2 \text{ mol}, \; n_{Cl^-} = 2 \times 0.2 = 0.4 \text{ mol}.$$

$$\text{Molarity of } Cl^- = \frac{0.4 \text{ mol}}{\frac{(26.2+92.8)}{2.38 \times 1000} \text{ lit}} = \frac{2.38 \times 400}{119.0} = \frac{\cancel{238}^{2} \times 4}{\cancel{119}} = 8 \text{ molar}.$$

$$\text{Molality of } Cl^- = \frac{0.4 \text{ mol}}{\frac{(92.8 + 2 \times 0.2 \times 18)}{1000} \text{ kg of water}} = \frac{400}{92.8+7.2} = \frac{400}{100.0} = 4 \text{ molal}.$$

If it is desired to construct the following galvanic cell:

$Ag(s) \mid Ag^+$ (saturated AgI) $\|$ Ag^+ (saturated AgCl, $x \times 10^{-4}$ M Cl^-) $\mid Ag(s)$

to have $E_{cell} = 0.102$ V, what should be the value of x to get $[Cl^-]$, which must be present in the cathodic half-cell to achieve the desired EMF. Given K_{sp} of AgCl and AgI are 1.8×10^{-10} and 8.1×10^{-17}, respectively. (2.303 RT/F = 0.06, log 2 = 0.3)

Ans : $x = 4$

Sol^n: $K_{sp}(AgCl) = 1.8 \times 10^{-10}$

$(AgI) = 8.1 \times 10^{-17}$

$Ag / Ag^+(c_1) \| Ag^+(c_2) / Ag$

$\cancel{Ag} \longrightarrow Ag^+(c_1) + \cancel{e^-}$

$Ag^+(c_2) + \cancel{e^-} \longrightarrow \cancel{Ag}$

$Ag^+(c_2) \longrightarrow Ag^+(c_1)$, $E^\circ_{cell} = 0$

$s^2 = K_{sp}$

$s = \sqrt{K_{sp}}$

$c_1 = \sqrt{K_{sp}}$

$= 9 \times 10^{-9}$ M

$[Ag^+][Cl^-] = K_{sp}$

$[Ag^+] = c_2 = \dfrac{1.8 \times 10^{-10}}{x \times 10^{-4}}$

$0.102 = 0 - \dfrac{0.06}{1} \log\left(\dfrac{\cancel{9}^{5} \times 10^{-9}}{\cancel{1.8} \times 10^{-10}} \times x \times 10^{-4}\right)$

$-1.7 = \log(x \times 50 \times 10^{-4})$

$\log(2 \times 10^{-2})$

$= \log(x \times 5 \times 10^{-3})$

$x = \dfrac{2 \times 10^{-2}}{5 \times 10^{-3}}$

$= \dfrac{20}{5} = 4$

$x = 4$

Ans : ≃ 2%

A strong current of trivalent gaseous boron passed through a silicon crystal decreases the density of the crystal due to part replacement of silicon by boron and due to interstitial vacancies created by missing Si atoms. In one such experiment, one gram of silicon is taken and the boron atoms are found to be 1000 ppm by mass when the density of the Si crystal decreases by 12%. The percentage of missing vacancies due to silicon, which are filled up by boron atoms nearly is (Atomic masses: Si = 30, B = 11)

10^6 g final —— = 1000 g 'B'

∴ 1000 g final crystal has 999 g Si, 1 g 'B'

→ Let us take 100 Silicon atoms in sample initially

→ After doping, x 'Si' atoms missing and y 'B' atoms doped.

→ Taking (assuming) volume of crystal to remain constant

So % decrease in density due to doping = % decrease in mass.

$$\frac{(100-x)}{\cancel{N_A}}\times 30+\frac{y}{\cancel{N_A}}\times 11 = 88\%\text{ mass of }100\text{ Si atoms} = 0.88\times\frac{100}{\cancel{N_A}}\times 30$$

$$3000-30x+11y = 2640 \Rightarrow 30x-11y = 360 \quad (1)$$

$$30x - 11y = 360 \quad \text{--- (1)}$$

$$\frac{\frac{(100-x)}{\cancel{N_A}} \times 30 \text{ g}}{\frac{y}{\cancel{N_A}} \times 11 \text{ g}} = \frac{999}{1}$$

In final crystal

$$3000 - 30x = 999 \times 11y$$

$$30x + 999 \times 11y = 3000 \quad \text{--- (2)}$$

$$(2) - (1) \quad 999 \times 11y + 11y = 2640$$

$$11y \times 1000 = 2640$$

$$\Rightarrow y = \frac{2.64}{11} = 0.24$$

$$x = \frac{360 + 2.64}{30} = \frac{362.64}{30}$$

% of 'Si' vacancies filled 'B' atoms

$$= \frac{y}{x} \times 100 = \frac{0.24}{362.64} \times 30 \times 100 = \frac{24 \times 30}{362.64} = \frac{720}{362.64} = 1.96\ldots \approx 2\% \text{ Ans.}$$